Day 1 — What is a Magic Square?

Module: Magic Squares (*Bhadragaṇita*) · **Band:** Middle · **Time:** 45 min

Learning objective

By the end of this lesson, students will (a) state the definition of a magic square, and (b) construct a 3×3 magic square with the numbers 1 to 9 by their own trial.

Materials

- Printed blank 3×3 grid (3 per student, with the digits 1–9 listed beside the first grid)
- A loose set of paper tiles or small cards numbered 1–9 (one set per pair)
- Whiteboard

Vocabulary introduced today

- *magic square* — a square grid in which every row, every column, and both main diagonals add up to the same total.
- *magic constant* — that common total (sometimes also called the *magic sum*).

Lesson flow

Warm-up (5 min)

- Teacher draws an empty 3×3 grid on the board. *"I want to place the numbers 1, 2, 3, 4, 5, 6, 7, 8, 9 in this grid — each number exactly once — so that* ***every row, every column, and both diagonals add up to the same number****. Can it even be done?"*
- Take 2–3 quick guesses. Don't confirm. Write on the side of the board: "Possible? If yes — what's the magic sum?"

Core (20 min)

1. **Hand out tiles 1–9 and the blank grids.** Pairs work for 8 minutes to try to build one.

2. **Mid-pair check-in (after 4 min):** *"What number is in the centre of your best attempt so far?"* Most pairs will have 5 in the centre. Ask why.

3. **Give a hint, not the answer.** *"The middle number is 5. Each row sums to 15. Now try again."*

4. **Reveal one valid square** on the board (after most pairs have struggled — 7–8 min in):

```
2 | 7 | 6
---------
9 | 5 | 1
---------
4 | 3 | 8
```

Verify together: each row → 15. Each column → 15. Both diagonals → 15. *"That sum 15 is called the* magic constant *of this square."*

5. **Definition on the board.** Students copy this into their notebooks:

   A **magic square** is an $n \times n$ grid filled with distinct numbers such that every row, every column, and both main diagonals add up to the same total — the **magic constant**.

Activity (15 min) — Verify and vary

- **Part A (5 min):** Each student picks any 3 lines (rows, columns, or diagonals) of the revealed square and verifies the sum on their notebook.
- **Part B (10 min):** Pairs try to make a *different* 3×3 magic square using 1–9. (Without saying so, you are previewing Day 3 — they will discover symmetries.) Most will produce rotations or reflections of the same square. That is the discovery, not the failure.

Wrap-up (5 min)

- Quick show of hands: "Did anyone find a square that is genuinely different from the one on the board?" (Most will say yes; some of those "different" squares will be rotations.)
- Preview Day 2: *"Tomorrow we'll prove why the magic constant is exactly 15 for the 3×3 with 1–9. We won't guess — we'll derive it."*

Student-facing content

A magic square is an $n \times n$ grid of distinct numbers where every row, every column, and both main diagonals add up to the same total — the **magic constant**.

The simplest non-trivial magic square uses the numbers 1 to 9 in a 3×3 grid. Its magic constant is **15**.

```
2 | 7 | 6
---------
9 | 5 | 1
---------
4 | 3 | 8
```

Verify: rows → 2+7+6=15, 9+5+1=15, 4+3+8=15. Columns → 2+9+4=15, 7+5+3=15, 6+1+8=15. Diagonals → 2+5+8=15, 6+5+4=15.

Homework

- On graph paper, copy the magic square shown in class. Then, beside it, draw three different "rotations" of it (turn the page 90°, 180°, 270°). Are they still magic squares? Verify one row of each.

Differentiation

- **One tier up:** Try to make a 4×4 magic square using the numbers 1 to 16. (Hard! The magic constant is 34. You will likely fail without a method. That's fine — we'll learn the method later.)

- **One tier down:** Practise just verifying the revealed 3×3 square. Sum every row, column, and diagonal slowly and carefully.

Teacher notes

- Resist the urge to give the answer in the first 5 minutes. The discovery is the lesson.
- Some students will ask "is it always 15?" Answer: "for THIS square, with THESE numbers (1–9). Tomorrow we'll see why."
- A few students will try to use the same number twice. Reinforce that all 9 numbers must be distinct — that's part of the definition.

Citation(s) used in this lesson

- Nārāyaṇa Paṇḍita, *Gaṇita-kaumudī*, Book 14 (*Bhadragaṇita*), 1356 CE — the historical source for the systematic treatment of magic squares we will follow this module. Today we set the stage; Day 4 begins his explicit method.